Linear Inequalities

Name _____ **Date** _____ No calculator Version 1

KEY WORDS inequality · greater than · less than · at least · at most · solve · number line

A Writing inequalities ●○○

ANSWER ON THE SHEET

Remember: $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

1. Write each sentence as an inequality.

a) x is greater than 4 _____

b) x is less than 9 _____

c) x is at least 2 _____

d) x is at most 10 _____

e) x is more than −1 _____

f) x is no more than 6 _____

g) x is no less than −5 _____

h) x is under 0 _____

2. Circle every number that makes the inequality true.

a) $x > 5$: 3, 5, 5.5, 9

b) $x \leq 2$: -1, 0, 2, 3

c) $x \geq -3$: -4, -3, -2.5, 0

d) $x < 0$: -2, -0.5, 0, 1

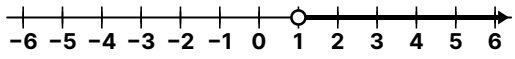
B Number lines ●○○

ANSWER ON THE SHEET

Remember: An open circle means the number is not included. A filled circle means it is included.

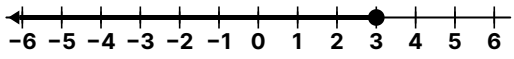
3. Write the inequality shown on each number line.

a)



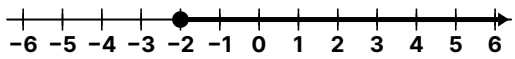
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b)



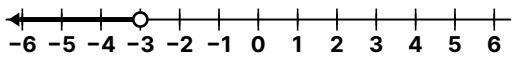
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c)



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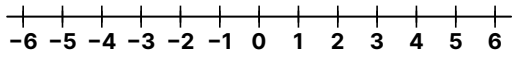
d)



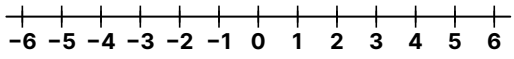
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4. Show each inequality on its number line.

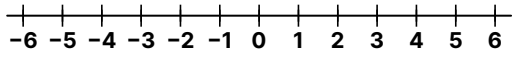
a) $x > -2$



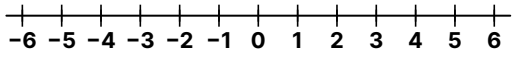
b) $x \leq 5$



c) $x \geq 0$



d) $x < 4$



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C Solving inequalities ●●○

WORK IN YOUR BOOK

Remember: Solve an inequality the same way as an equation. Keep the inequality sign in every line.

5. Solve.

a) $x + 4 > 9$

b) $x - 3 \leq 3$

c) $3x < 12$

d) $5x \geq 15$

e) $2x + 1 > 9$

f) $3x - 2 \leq 13$

g) $4x + 5 < 13$

h) $2x - 7 \geq 5$

i) $\frac{x}{2} > 5$

j) $\frac{x}{3} \leq 4$

k) $6x + 1 > -5$

l) $5x - 4 < -4$

m) $2(x + 1) > 10$

n) $3(x - 2) \leq 12$

o) $\frac{x}{4} + 1 \geq 3$

D Dividing by a negative ●●●

WORK IN YOUR BOOK

Watch out: When you multiply or divide both sides by a negative number, turn the inequality sign around.

6. Solve.

a) $-2x > 8$

b) $-x \leq -3$

c) $-3x \geq -6$

d) $-5x < 5$

e) $6 - x \leq 4$

f) $9 - 2x > 1$

g) $-2x + 3 < 9$

h) $-4x - 1 \geq 7$

i) $1 - 3x < 7$

★ Extension

FOR EARLY FINISHERS

7. Aroha has \$40. She pays \$4 to get into a fair, and each ride costs \$6. Write an inequality for the number of rides x she can go on, and solve it. What is the largest number of rides?

8. The solution of an inequality is $x > 2$. Which of the whole numbers 0, 1, 2, 3, 4 and 5 are solutions?

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Linear Inequalities

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS inequality · greater than · less than · at least · at most · solve · number line

A Writing inequalities ●○○ ANSWER ON THE SHEET

Remember: $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

1. Write each sentence as an inequality.

a) x is greater than 6 _____

b) x is less than 7 _____

c) x is at least 3 _____

d) x is at most 12 _____

e) x is more than −2 _____

f) x is no more than 8 _____

g) x is no less than −4 _____

h) x is under 0 _____

2. Circle every number that makes the inequality true.

a) $x > 4$: 2, 4, 4.5, 7

b) $x \leq 3$: -2, 0, 3, 4

c) $x \geq -2$: -3, -2, -1.5, 0

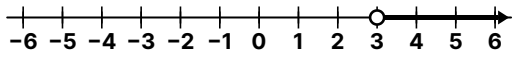
d) $x < 1$: -3, 0.5, 1, 2

B Number lines ●○○ ANSWER ON THE SHEET

Remember: An open circle means the number is not included. A filled circle means it is included.

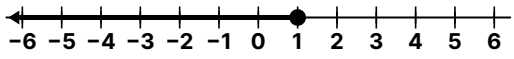
3. Write the inequality shown on each number line.

a)



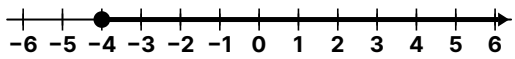
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b)



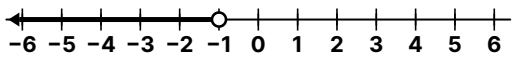
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c)



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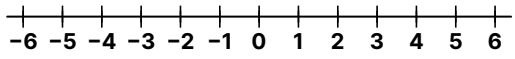
d)



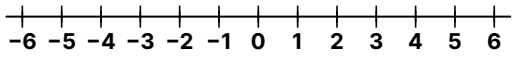
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4. Show each inequality on its number line.

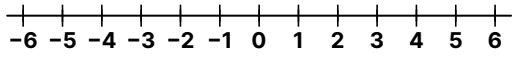
a) $x > -4$



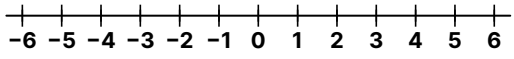
b) $x \leq 3$



c) $x \geq 1$



d) $x < 2$



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C Solving inequalities ●●○ WORK IN YOUR BOOK

Remember: Solve an inequality the same way as an equation. Keep the inequality sign in every line.

5. Solve.

a) $x + 6 > 9$ **b)** $x - 5 \leq -1$ **c)** $4x < 12$

d) $2x \geq 14$ **e)** $3x + 1 > 7$ **f)** $2x - 5 \leq 7$

g) $5x + 3 < 13$ **h)** $3x - 8 \geq 4$ **i)** $\frac{x}{3} > 3$

j) $\frac{x}{2} \leq 7$ **k)** $4x + 3 > -1$ **l)** $6x - 5 < -5$

m) $2(x + 3) > 10$ **n)** $4(x - 1) \leq 12$ **o)** $\frac{x}{5} + 2 \geq 4$

D Dividing by a negative ●●● WORK IN YOUR BOOK

Watch out: When you multiply or divide both sides by a negative number, turn the inequality sign around.

6. Solve.

a) $-3x > 9$ **b)** $-x \leq -5$ **c)** $-2x \geq -8$

d) $-4x < 8$ **e)** $7 - x \leq 4$ **f)** $11 - 2x > 5$

g) $-3x + 2 < 8$ **h)** $-5x - 1 \geq 4$ **i)** $2 - 3x < 5$

★ Extension FOR EARLY FINISHERS

7. Tane has \$45. He pays \$5 to get into a fair, and each ride costs \$8. Write an inequality for the number of rides x he can go on, and solve it. What is the largest number of rides?

8. The solution of an inequality is $x > 3$. Which of the whole numbers 0, 1, 2, 3, 4 and 5 are solutions?

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